Opinion Network Formation: Three Readings of One Class Survey

Team: A Shortest Path to an A Grade

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Dynamical Processes in Complex Networks - Assignment 1

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Code Repository: https://github.com/Fresh-butter/DPCN_Assignment_1

The problem this survey poses

About 80% of this class agrees with almost everything. Out of 5158 answers actually given across 60 statements, 4127 are "Agree" or "Strongly Agree", and per-item means sit between 3.5 and 4.5 on a five-point scale. If you just take the raw responses and compute cosine similarity between students, everyone looks nearly identical ($\sim$0.96 for every pair), and you end up with a near-complete graph that tells you nothing.

So the real question isn't what people answered. It's how far each answer sits from what everyone else said. Once we centre each item on its mean, genuine structure shows up, and there turns out to be quite a lot of it.

We built three networks from this one dataset. They use the same cleaned data but differ in what counts as a node and what counts as a relationship, so they end up answering different questions. The most interesting results come from putting them side by side: where two of them agree, it's hard to dismiss as an artifact, and where they disagree, the disagreement itself usually turns out to be interesting.

1 Dataset documentation

The raw file is 96×61: one respondent identifier and 60 Likert statements, split into four blocks of fifteen, i.e., **T**echnology, **E**ducation, **S**ociety/Ethics, and en**V**ironment. All cleaning is done once in `common/contract.py` and imported by all three members, so nobody re-cleans the data separately.

Step	Result	Rationale
Encode Likert 1–5	numeric	Strongly Disagree = 1 ... Strongly Agree = 5.
"No Comments"/blank $\rightarrow$ NaN	-	We didn't encode these as 3 (Neutral). Treating a non-answer as "Neutral" would fabricate data right at the centre of the scale and push correlations toward zero.
Drop rows with >30 of 60 missing	$96 \rightarrow 87$	Natural break in the data (see below).
Respondents $\times$ items matrix	87×60	62 missing cells remain.
Per-item z-score, computed on all 87	-	This is what makes the rest of the analysis work; see below.

Why 30. There's a natural gap in the data. 68 rows have zero missing answers, another 17 have 1–4, then one row each has 10 and 15. After that the count jumps straight to 45 (four rows) and 60 (five rows). Nothing at all falls between 15 and 45. Any threshold in that gap keeps exactly the same 87 respondents. The nine rows we drop are basically abandoned responses, not partial ones.

Why z-score. This is what makes everything else possible. Raw pairwise correlation between students averages 0.267; after centring each item on its own mean, pairwise-complete cosine similarity averages -0.010 and spreads across $[-0.638, +0.758]$. The ceiling goes away and actual variation becomes visible. We z-score over all 87 respondents so that the three members' values are on the same reference frame.

What we didn't standardise. The 62 remaining missing cells are handled differently by each construction, since that's really a modelling choice: pairwise-complete cosine with $\geq$20 shared items (A), pairwise-complete Pearson with `min_periods=10` (B), and simply no edge for a missing cell (C). Each member also ran a mean-imputed variant as a sanity check; the differences are tiny (for B, mean $|\Delta r| = 0.0033$).

Two items, E02 and T10, looked like they might be reverse-worded. We flagged them but didn't flip them. Since we have a signed network, it should be able to discover reverse-coding as negative edges on its own. E02 does end up behaving as a minority-opinion item; T10 does not.

Because all three members import the same loader, the three pipelines produce bit-identical 87×60 matrices. This is what lets us do the cross-network comparisons in Section 4: A and C describe the same 87 students, B and C the same 60 statements, so every comparison is one population measured two ways.

2 Pipeline

	Nodes	Edge rule	Edges	Question it answers
A Agreement	87 students	cosine of z-vectors > 0.15	822	Who thinks alike?
B Signed statement	60 statements	Pearson r, permutation, FDR	402	Which opinions move together or clash?
C Bipartite deviation	87 students, 60 items	student–item edge, $\lvert z\rvert > 1$	1200	Who breaks consensus, on what?

A: agreement network. Two students get an edge when their z-scored response vectors are similar enough. A cosine threshold of 0.15 gives 822 edges, mean degree 18.9, and one connected component. This isn't a knife-edge choice: sweeping the threshold, the giant component holds every node up to 0.225 and only drops below half past 0.325. We also built a mutual-kNN version ($k = 8$) as an alternative, which we use for the spreading dynamics in Section 3.4.

B: signed statement network. Statements are connected by the Pearson correlation of their response profiles across students, keeping the sign so that opposition shows up as negative edges. We test significance against a permutation null where each item is independently shuffled, with Benjamini–Hochberg FDR at $q = 0.05$. 402 of 1770 candidate pairs survive (smallest retained $|r|$ is 0.274). We first tried a family-wise max-statistic threshold, but that demanded $|r| > 0.50$ and killed every negative edge- which defeated the whole point of building a signed network. The per-edge FDR rule keeps 391 positive and 11 negative edges.

C: bipartite deviation network. Instead of removing the ceiling effect and studying what's left, this one makes the residual the actual object of study. A student gets connected to a statement when they answered it more than one SD from the class mean, weighted by $|z|$. At cutoff 1.0 this gives 1200 edges, with every student and every statement participating.

All these thresholds are somewhat arbitrary, i.e. a slightly different value would also be defensible. So each one was swept. The short version: coarse structure doesn't change, fine detail does. A's threshold and kNN rules share only 25.2% of their edges by Jaccard overlap but still give the same two communities, and C's cutoff range 0.75–1.5 swings the edge count from 2856 to 442 without changing any of the conclusions below.

All results use `SEED = 42`. We also checked that re-running everything from scratch produces identical outputs.

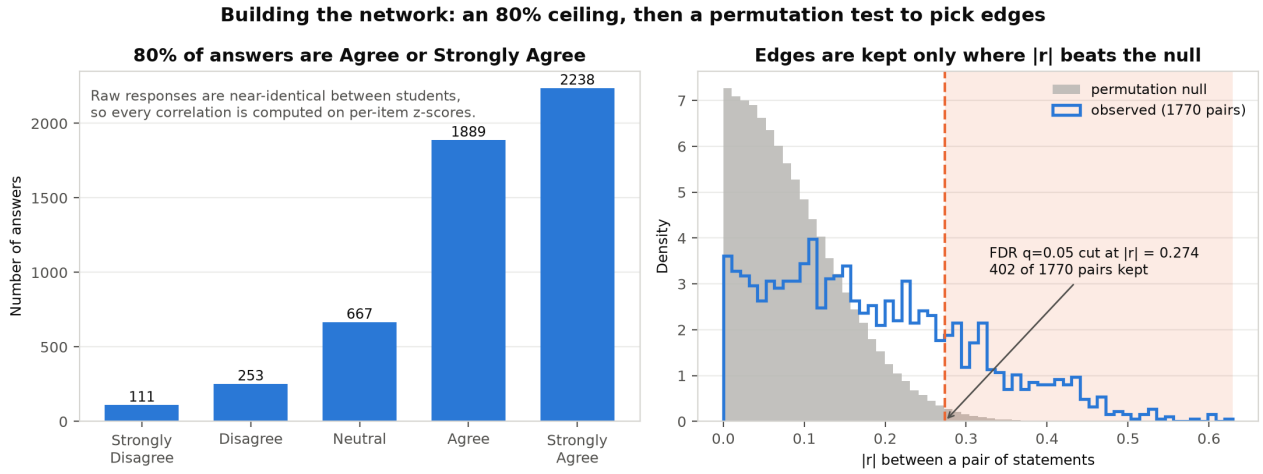


Figure 1: The problem and one response to it. *Left:* the response distribution, i.e. 80% of answers given are Agree or Strongly Agree, which is why every construction in this report operates on z-scores rather than raw responses. *Right:* the observed distribution of $|r|$ across all 1770 statement pairs sits clearly to the right of a permutation null, and the FDR cut at $|r| = 0.274$ retains exactly the part that separates.

3 Analysis and visualisations

3.1 The class splits into two blocs, and it's about intensity

Louvain on the agreement network finds two main communities of 48 and 35 students (plus two tiny pairs) at modularity $Q = 0.432$. On its own that number doesn't mean much. Louvain will find some partition on any graph. What matters is the comparison to a degree-preserving configuration model, which scores 0.178 ± 0.007 over 30 rewirings. The observed value is **35.5 standard deviations above that**. The partition is also strongly assortative: 746 of 822 edges (90.8%) stay within a community, and within-community edges carry more weight than between-community ones (0.286 vs 0.212).

The network is also much more transitive than you'd expect by chance. Average local clustering is 0.505 against an Erdős–Rényi null of 0.220 ± 0.009, about 32 SD above. If a student agrees with two others, those two very likely agree with each other too. Average shortest path length (2.054) is actually slightly *longer* than the null (1.792), so the network is locally cliquey rather than small-world.

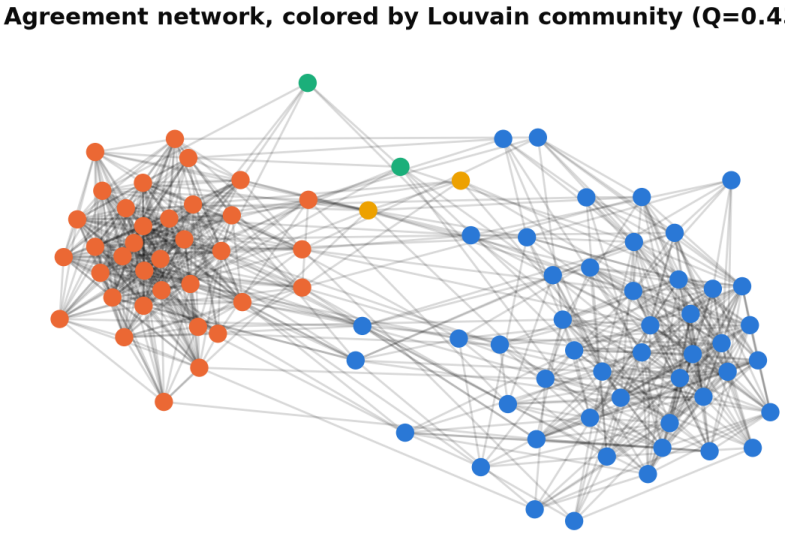


Figure 2: The agreement network over 87 students, coloured by Louvain community. The two main blocs are clearly visible and far from random ($Q = 0.432$ vs a configuration null of 0.178 ± 0.007).

The more interesting part is *what* these two blocs actually differ on. They're not opposed camps. Both rank the same statements highest; they just differ in how strongly they agree. One bloc (35 students) has overall mean z of $+0.371$, the other (48 students) -0.283. It's a split in conviction, not in position. One group feels strongly, the other is more moderate, but they're not on opposite sides.

And this intensity gap is concentrated in specific topics. Of the twenty statements that best separate the blocs, eleven are Environment and seven are Society/Ethics; only one each from Technology or Education. Mean absolute z-gap per item by block: V 0.986, S 0.835, E 0.536, T 0.434.

3.2 Disagreement is rare, but it's concentrated

The signed statement network is overwhelmingly positive: 391 of 402 edges. The eleven negative edges are the interesting ones, and they're not scattered randomly. Eight of the eleven touch a single statement - E03 ("Class attendance should be compulsory for all courses"), which anti-correlates with a cluster of progressive-education and innovation items (E04, E07, E10, E11, E12, T06, S06). What looks like random dissent is actually one clear axis: traditionalist vs progressive views on pedagogy.

The network is also almost perfectly balanced. Out of 1298 closed triads, **zero** are frustrated; every existing triangle has consistent signs. The frustration index (minimum sign flips needed for a clean two-faction split) is 2 out of 402 edges. A null model that reshuffles which edges carry the eleven negative signs while keeping the topology fixed gives 10.98 ± 0.16, putting the observed value 56 SD *below* chance. (The two "wrong" edges must sit on a longer signed cycle since the triangle census is already perfect.) Basically, whatever this class believes, it believes it consistently. There are almost no cases of holding contradictory positions.

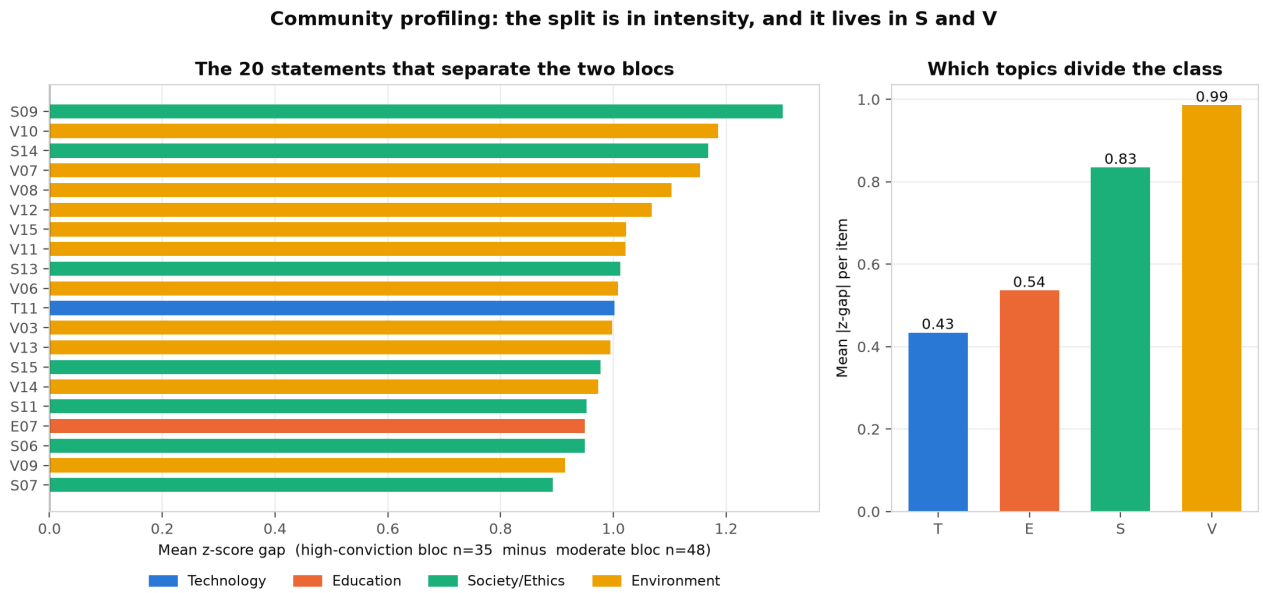


Figure 3: What separates the two blocs. *Left:* the twenty most discriminating statements, Eighteen of them are Environment or Society/Ethics. *Right:* the same result aggregated by block. The class divides over how strongly to hold environmental and social positions, not over technology or education.

One more thing: Louvain on the positive backbone finds eight communities whose match with the survey's T/E/S/V blocks is weak (NMI = 0.277). The class doesn't organise its opinions along the topic boundaries the survey designers set up. The clearest example is T08 (AI-assisted diagnosis) and T09 (autonomous vehicles), which survive the FDR cut with *zero edges*. Two Technology statements that don't correlate with anything in the survey, including the rest of Technology.

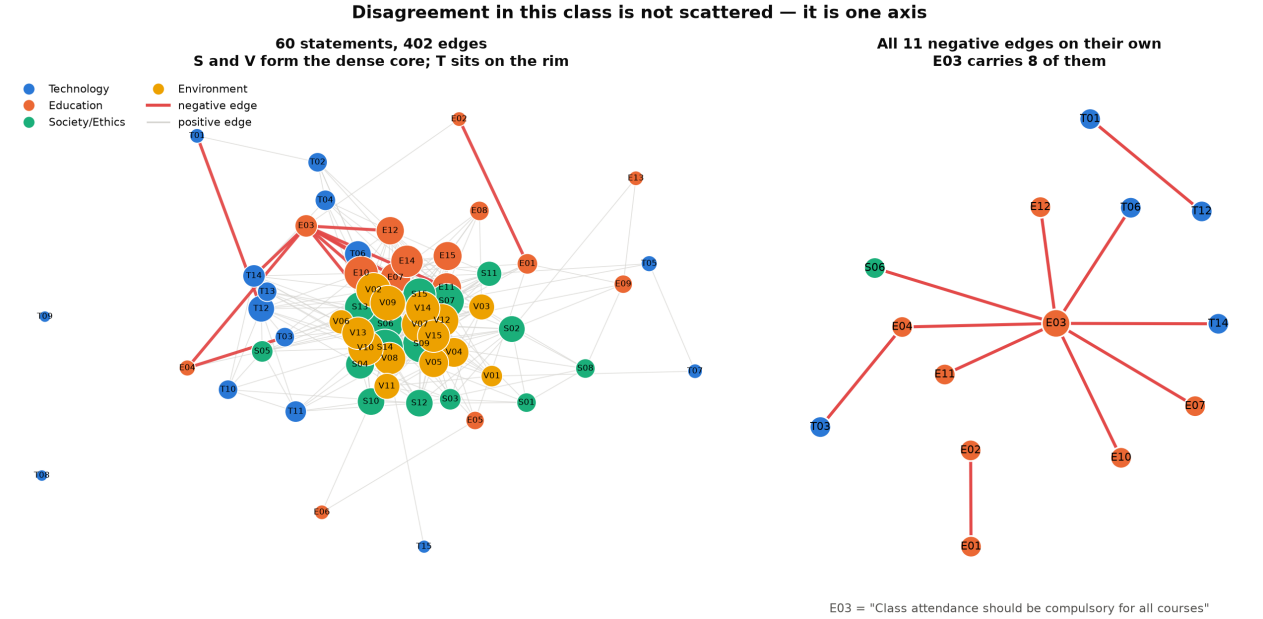


Figure 4: The signed statement network. *Left:* Society and Environment statements form a dense core while Technology sits on the rim, with T08 and T09 fully isolated. *Right:* all eleven negative edges drawn alone; E03 carries eight of them.

3.3 Consensus has structure; dissent does not

The bipartite network asks the opposite question, i.e., not who agrees, but who breaks from the class and on what. The most deviated-on statement is T13 (cybersecurity investment), where 60 of 87 students answered more than one SD from the mean. Interestingly, the lowest-mean items are *not* the highest-deviation items:

E03 and E02, which are central to B's disagreement structure, rank only eleventh and below here. A low mean means the class agrees something is false; high deviation degree means the class can't settle on it. Those are different things.

Projecting this bipartite graph onto the 87 students, i.e., connecting two students when they deviate on the same issues, gives what was probably our most surprising result. Louvain returns two communities at $Q = 0.0368$, which is low. But low isn't the same as meaningless, so we compared it against a degree-preserving null, same as we did for network A. The null scores 0.0567 ± 0.0032. So the co-dissent network isn't just weakly clustered, it's **6.1 SD *less* clustered than a random graph** with the same degree sequence.

Compare that with A's +35.5 SD on the same 87 students, and you get the main finding: when students are linked by what they agree on, they form two clear camps. When the exact same students are linked by what they dissent on, there's basically no structure at all. There's no "contrarian clique", disagreeing with the class on one thing doesn't predict what else you'll disagree on.

Sorting the incidence matrix by degree does give a nested-looking, triangular pattern, and NODF = 41.8 suggests a hierarchy of dissent, i.e. that students work down a shared ordering of how contentious each statement is. But that doesn't hold up against a null either. NODF goes up automatically with heterogeneous degrees, and degrees here range from 3 to 35 deviations per student. Against a curveball null that fixes both degree sequences, the null NODF is 41.00 ± 0.26: the observed value clears it by just +0.79 points out of 100 ($z = +3.08$). Technically above chance, but practically nothing.

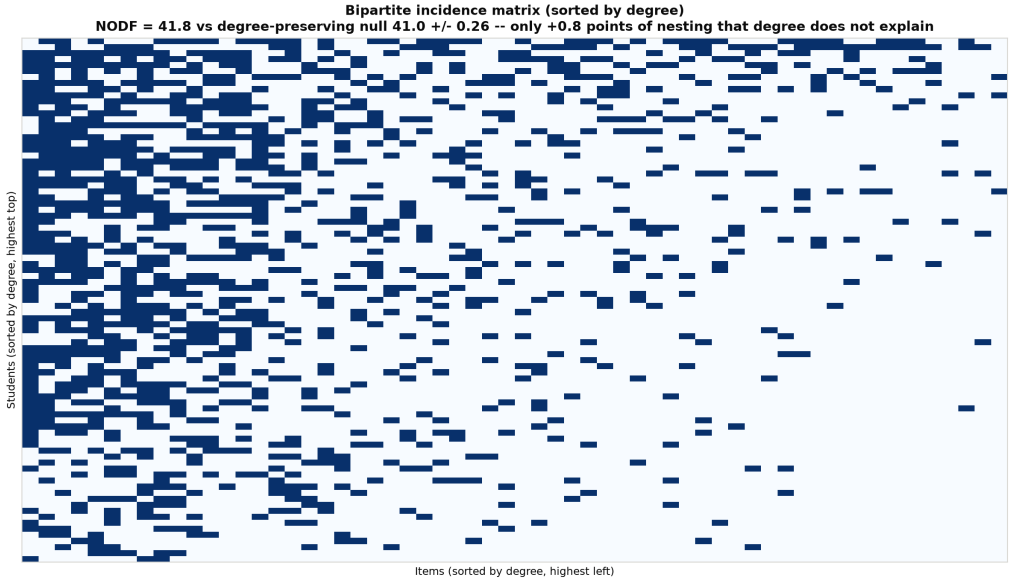


Figure 5: The bipartite incidence matrix, students by statements, sorted by degree. The nested triangular shape is real but almost entirely explained by the degree sequence: only +0.8 NODF points out of 100 survive a null that fixes both degree sequences.

3.4 Running processes on these networks

In a few places, running a dynamical process on the network showed things that the static structure alone didn't.

Random walk on the deviation network is all about degree. A walker going student → item → student can only travel between students through an issue they both deviate on, so mean first-passage time (MFPT) works as an issue-mediated distance. Computed exactly from the fundamental matrix with the true stationary distribution $\pi_j \propto \text{strength}_j$, MFPT ranges from 39.8 to 1063.5 steps, i.e. a 27-fold spread that looks like it should mean something interesting. But it doesn't, really. MFPT correlates with deviation degree at Spearman $\rho = -0.971$, and dividing out node strength leaves a relative MFPT range of just 1.005–1.118. The deviation network is basically indistinguishable from a configuration model: once you know how often a student deviates, you know pretty much everything about how the dynamics reach them. This is the same thing the modularity and nestedness nulls told us, just from a third angle.

Seeding high-centrality students backfires on sparse graphs. We ran SI spreading from three seeds, 30 trials per condition, comparing top-eigenvector seeds against random ones. On the dense threshold graph

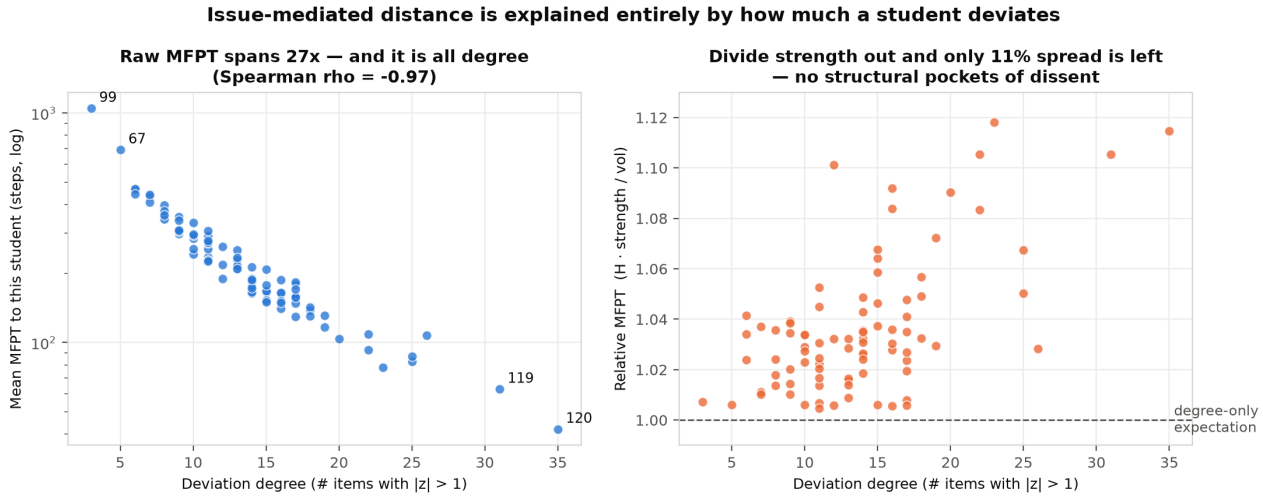


Figure 6: Mean first-passage time on the bipartite deviation network. *Left:* raw MFPT spans 27-fold, but tracks deviation degree at $\rho = -0.971$. *Right:* dividing out node strength leaves only 11% residual spread.

the choice doesn't matter: both reach basically the whole class. On the sparser mutual-kNN graph, centrality seeding is actually *worse*: coverage at $t = 30$ is 0.244 ± 0.266 for the top eigenvector seeds vs 0.487 ± 0.401 for random (mean $\pm$ SD, 30 trials; Welch $t = 2.76$, $p = 0.008$). Worth noting is that on a graph this sparse an outbreak either takes off or dies out, so the SD is as large as the mean and individual trials range from near-zero to near-complete coverage.

The reason is seed redundancy. Eigenvector centrality rewards being in the same dense core, so the three top students are all right next to each other. Their mean pairwise distance 1.00 hops, versus 3.33 for random seeds on the same graph. They're basically covering the same area. On the dense graph where everyone is about 2 hops from everyone else, that redundancy doesn't cost anything. But make the graph sparser and it becomes the dominant effect.

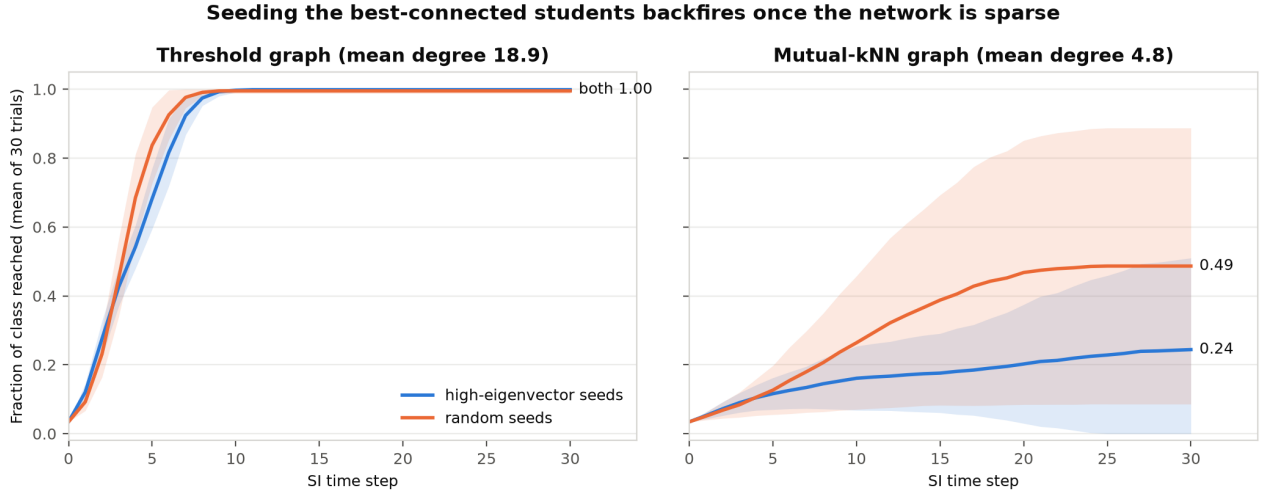


Figure 7: SI spreading from three seeds, 30 trials per condition, shading ± 1 SD. *Left:* on the dense threshold graph the seeding strategy makes no difference. *Right:* on the sparse mutual-kNN graph random seeding reaches roughly twice as many students on average, though the bands overlap a lot; on a sparse graph results are close to all-or-nothing.

We also ran Deffuant–Weisbuch bounded-confidence dynamics on the agreement network and Altafini signed consensus on the statement network. Both behaved as expected. A fragmentation-to-consensus transition around $\varepsilon \approx 0.30$–0.55, and slow monotone decay from 1.21 to 0.08 (consistent with a nearly balanced network). Neither gave us anything that distinguishes this class from any other, so we just note them here briefly.

4 Results and discussion

4.1 Two different constructions rank the same statements the same way

Member A measures, for each statement, the gap in mean z-score between the two student blocs, i.e. how much that statement separates the class. Member B measures how far a perturbation to that statement propagates through the signed correlation network, its dynamical leverage. These come from different graphs, different node sets, and different math. Across the 58 statements both can score, they agree at Spearman $\rho = +0.784$ ($p = 3.5 \times 10^{-13}$; Pearson $r = +0.617$).

Grouped by block, they give the same ordering: **V** > **S** > **E** > **T**.

Block	A: mean |z-gap|	B: mean leverage	C: mean deviation degree
Environment (V)	0.986	0.957 ± 0.014	14.13
Society/Ethics (S)	0.835	0.933 ± 0.054	19.33
Education (E)	0.536	0.762 ± 0.235	23.13
Technology (T)	0.434	0.714 ± 0.249	23.40

The spread is also worth looking at. Environment's leverage has an SD of 0.014, i.e. all fifteen V statements fall between 0.914 and 0.967. Technology's is eighteen times larger at 0.249, ranging from 0.233 to 0.953. Environment acts as one tightly coupled bloc in the opinion graph; Technology is more of a loose collection of statements that just happen to share a label. This lines up with B finding that the class's communities don't match the survey's topic blocks, and with T08 and T09 not correlating with anything at all.

4.2 Influence and deviation go in opposite directions

The third column in that table runs backwards. Member C's deviation degree, how many students answered a statement more than one SD from the mean, anti-correlates with both other measures, i.e. $\rho = -0.390$ ($p = 0.002$) against B's leverage, and $\rho = -0.349$ ($p = 0.006$) against A's bloc gap.

So the statements with the most influence and the sharpest class-dividing power are the ones individuals deviate from *least*. Environment is where the class is most tightly coupled. V statements sort students into camps by *how strongly* they agree, while attracting almost no individual dissent. Technology and Education are where people actually break ranks.

This also means there are two different senses of "controversial" here, and they point in opposite directions. B finds E03 as the disagreement axis (8 of 11 negative edges). C finds T13 as the biggest flashpoint (60 of 87 students deviating). Both are correct. E03 produces *patterned anti-correlation*, splitting the class into opposed sides. While T13 produces *raw spread*, where everyone answers differently but with no particular pattern.

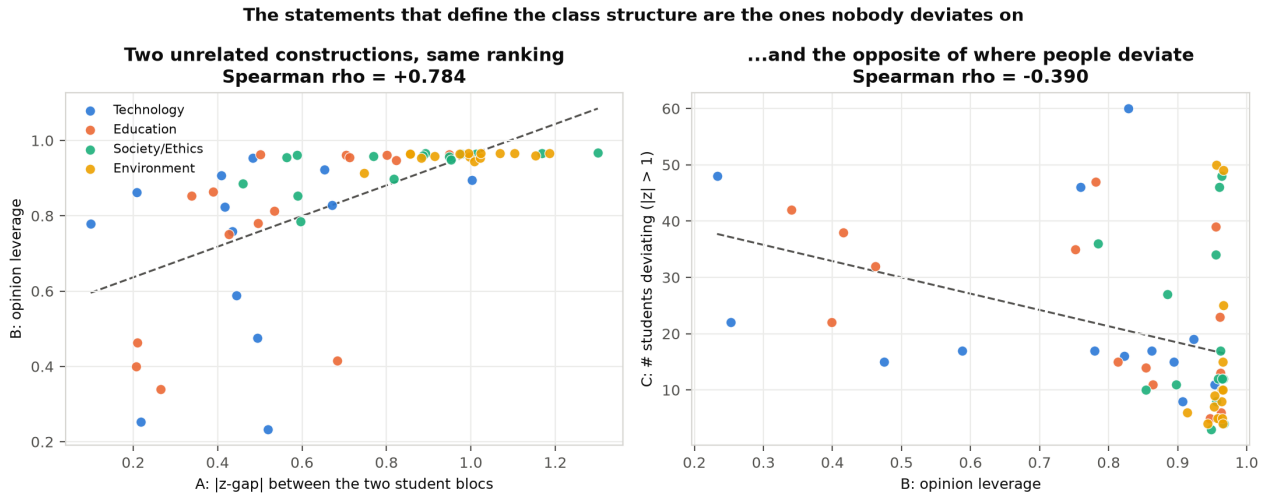


Figure 8: *Left:* two independent constructions rank the same 58 statements almost identically. *Right:* both run opposite to how much individual students deviate on a statement. Note that leverage saturates near 0.96, so the positive relationship is driven mostly by the lower tail.

4.3 Putting the three networks together

The clearest overall result is the asymmetry between agreement and dissent. On the same 87 students, agreement structure sits 35.5 SD above its null while co-dissent sits 6.1 SD *below* its own. Three independent measurements on network C, modularity, nestedness, MFPT, all say the same thing, i.e. once you know how often a student deviates, you know basically everything about where they sit in the network.

That agreement and dissent really are measuring different things (not just two views of one structure) is confirmed directly. Correlating A's pairwise similarity against C's symmetrised issue-mediated distance across 3741 student pairs gives $r = +0.036$ ($\rho = +0.026$), 0.1% of variance explained. The p-value only clears 0.05 because 3741 pairs make tiny effects detectable; there's essentially no relationship here. Knowing two students think alike tells you nothing about whether they break from the class on the same questions.

Stepping back, the picture of this class is that a cohort that agrees on most things and agrees consistently, clustering at 2.3× the random baseline, zero frustrated triangles out of 1298. The real division isn't topical but about intensity, separating those who feel strongly about environmental and social issues from those who hold the same views more mildly. The disagreements that exist are concentrated on a single pedagogical axis rather than spread across topics. And individual dissent is genuinely idiosyncratic; students break from consensus at very different rates, but *which* issues they break on seems to be personal rather than factional.

4.4 Limitations

This is a single class of 87 students, so none of this generalises beyond it. The ceiling effect, while handled, still limits what we can claim: after z-scoring we're analysing residuals around a near-universal agreement, and the "moderate" bloc still agrees with most statements. Every network needed a threshold, and while the overall structure is robust to all of them, individual edges aren't. A's two edge rules share only a quarter of their edges. Claims about specific edges or individual students should be taken with that in mind; claims about blocs and null comparisons hold across the sweeps. Also, leverage saturates near 0.96, so the $\rho = +0.784$ correlation in Figure 8 is driven disproportionately by the lower tail.

5 Individual Contribution

The project was divided into three main network analyses, with each team member taking the lead on one. All data cleaning and shared utilities were developed collaboratively in the `common/` module to ensure consistency across our analyses.

Member	Contribution
Vishal	**Agreement Network (`member_a`):** Constructed the student agreement network. Conducted community detection, structural analysis, centrality profiling, and simulated dynamical processes.
Harith	**Statement Network (`member_b`):** Built the signed statement network. Analysed network balance, identified hubs and isolates, compared community structures, and implemented Altafini dynamics.
Navneet	**Bipartite Network (`member_c`):** Developed the bipartite deviation network. Analysed nestedness, extracted one-mode projections, and evaluated structural robustness using Mean First Passage Time (MFPT).
Joint	**Shared & Cross-Analysis:** Developed `common/` utilities, conducted cross-network comparisons, and co-wrote the final report.